

# Schaffer Diaphragm Pressure Gauge (SH)

## Introduction

The Schaffer (SH) model process pressure gauges are designed to meet the stringent demands of low pressure measurements in tough conditions; they are built for high corrosive resistance and for a long product life.

The use of corrosion resistant material such as high grade stainless steel for both the casing and the measurement system, makes this pressure gauge best suitable to operate in an adverse environment and with corrosive media.

The welded design coupled with a modular construction system allows for best-in-class ingress protection (upto IP-65 standard). The gauge has a long and stable service life with tried and tested safety features such as vent plug and blow-off disc adhering to the EN 837-3 standard.



## Size and Range

|                |                             |
|----------------|-----------------------------|
| Nominal sizes: | 4"(100 mm) and 6"(150 mm)   |
| Ranges:        | 0 – 160 mmWc upto 6000 mmWc |

## Special Features

Compliance to latest EN837-3

Standard ranges : 0 – 160 mmWc upto 6000 mmWc

Overpressure limit - 130 % of FSD

Ingress protection to IP65

Accuracy class  $\pm 1.6\%$  of FS

Stainless steel casing

Socket to case - TIG argon welded

## Industrial Applications

Chemical and fertilizer

Petrochemical

Thermal, nuclear and hydroelectric power stations

Pharmaceutical

Food process

Water treatment plants

Iron and steel

Rubber molding / processing plants

Hydraulics and pneumatics

Sugar

Refineries

Hydraulic power packs

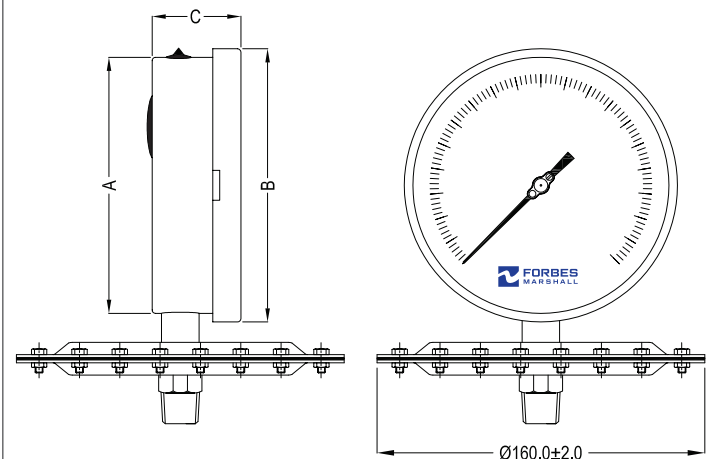
Cryogenics

| Standard Product Specifications |                                                       |
|---------------------------------|-------------------------------------------------------|
| Standards                       | Description                                           |
| Reference standard              | EN 837 - 3                                            |
| Mounting                        | Direct bottom                                         |
| Connection                      | ½" NPT (M)                                            |
| Case and bezel                  | SS 304                                                |
| Diaphragm                       | SS 316L                                               |
| Socket                          | SS 316L                                               |
| Movement                        | SS 304                                                |
| Top chamber                     | SS 316                                                |
| Bottom chamber                  | SS 316                                                |
| Dial                            | Aluminum, white background with black marking         |
| Pointer                         | Aluminum, black color micrometer with zero adjustable |
| Accuracy                        | ±1.6% of FS (as standard)                             |
| Overpressure limit              | As per EN standard                                    |
|                                 | 130% of maximum scale value                           |
| Window                          | Toughened glass                                       |
| Protection                      | IP65                                                  |
| Blow off disc and filling plug  | Neoprene                                              |
| Gasket                          | Silicon                                               |
| Ambient temperature             | -25°C to +65°C                                        |
| Process temperature             | Max. 65°C                                             |

| Standard Pressure Range Table |                 |      |                 |
|-------------------------------|-----------------|------|-----------------|
| Code                          | (Code – X) mmWc | Code | (Code – X) mmWc |
| AO                            | 0 + 250         | DU   | -250 + 0        |
| AP                            | 0 + 400         | DV   | -400 + 0        |
| AQ                            | 0 + 600         | DW   | -600 + 0        |
| AY                            | 0 + 1000        | DX   | -200 + 400      |
| AZ                            | 0 + 1600        | DY   | -100 + 150      |
| BA                            | 0 + 2500        | DZ   | -400 + 600      |
| BB                            | 0 + 4000        | EM   | -600 + 400      |
| BC                            | 0 + 6000        | EQ   | -500 + 500      |
|                               |                 | ES   | -1000 + 0       |

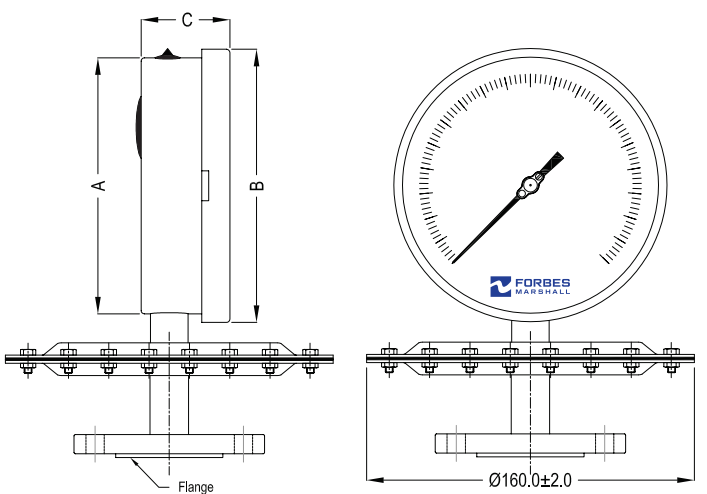
Mounting Options

Direct Bottom



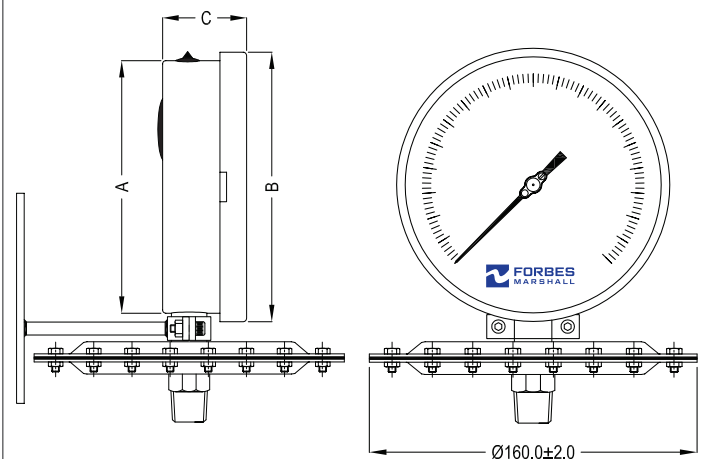
| Parameters<br>↓ | → Dial Size |      |
|-----------------|-------------|------|
|                 | 100         | 150  |
| A               | Ø100        | Ø150 |
| B               | Ø110        | Ø160 |
| C               | 48.5        | 50.0 |

Direct Bottom



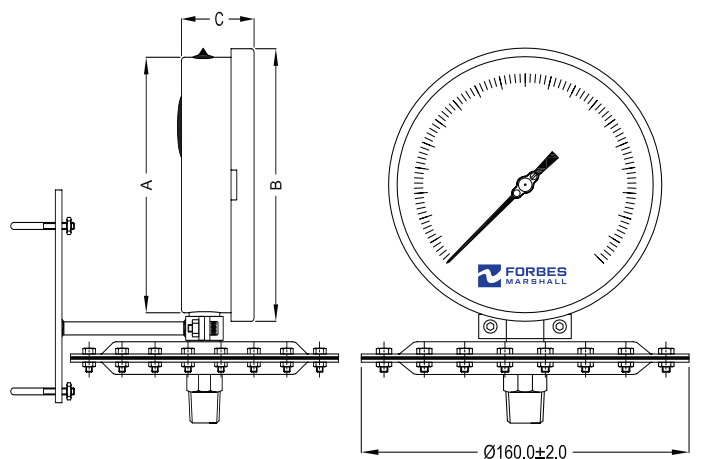
| Parameters<br>↓ | → Dial Size |      |
|-----------------|-------------|------|
|                 | 100         | 150  |
| A               | Ø100        | Ø150 |
| B               | Ø110        | Ø160 |
| C               | 48.5        | 50.0 |

Surface Bottom



| Parameters<br>↓ | → Dial Size |      |
|-----------------|-------------|------|
|                 | 100         | 150  |
| A               | Ø100        | Ø150 |
| B               | Ø110        | Ø160 |
| C               | 48.5        | 50.0 |

2" Pipe Mounting



| Parameters<br>↓ | → Dial Size |      |
|-----------------|-------------|------|
|                 | 100         | 150  |
| A               | Ø100        | Ø150 |
| B               | Ø110        | Ø160 |
| C               | 48.5        | 50.0 |

## Ordering Information

|                                     |    |    |   |   |   |   |    |   |           |  |  |  |  |
|-------------------------------------|----|----|---|---|---|---|----|---|-----------|--|--|--|--|
| <b>Model</b>                        | SH | ** | * | * | * | * | ** | * | * * * * * |  |  |  |  |
| <b>Dial Size</b>                    |    |    |   |   |   |   |    |   |           |  |  |  |  |
| 100 mm                              | E  |    |   |   |   |   |    |   |           |  |  |  |  |
| 150 mm                              | F  |    |   |   |   |   |    |   |           |  |  |  |  |
| <b>Mountings</b>                    |    |    |   |   |   |   |    |   |           |  |  |  |  |
| Direct Bottom                       | 1  |    |   |   |   |   |    |   |           |  |  |  |  |
| <b>Connection Size</b>              |    |    |   |   |   |   |    |   |           |  |  |  |  |
| 1/4"                                | A  |    |   |   |   |   |    |   |           |  |  |  |  |
| 3/8"                                | B  |    |   |   |   |   |    |   |           |  |  |  |  |
| 1/2"                                | C  |    |   |   |   |   |    |   |           |  |  |  |  |
| M20X1.5P                            | 1  |    |   |   |   |   |    |   |           |  |  |  |  |
| Specials                            | Z  |    |   |   |   |   |    |   |           |  |  |  |  |
| <b>Connection Type</b>              |    |    |   |   |   |   |    |   |           |  |  |  |  |
| NPT (M)                             | A  |    |   |   |   |   |    |   |           |  |  |  |  |
| BSP (M)                             | B  |    |   |   |   |   |    |   |           |  |  |  |  |
| BSPT (M)                            | C  |    |   |   |   |   |    |   |           |  |  |  |  |
| Metric Thread (M)                   | M  |    |   |   |   |   |    |   |           |  |  |  |  |
| Specials                            | Z  |    |   |   |   |   |    |   |           |  |  |  |  |
| <b>Range</b>                        |    |    |   |   |   |   |    |   |           |  |  |  |  |
| Refer to Std. Pressure Range Table  |    |    |   |   |   |   |    |   |           |  |  |  |  |
| <b>Engineering Unit*</b>            |    |    |   |   |   |   |    |   |           |  |  |  |  |
| mmWC                                | X  |    |   |   |   |   |    |   |           |  |  |  |  |
| <b>Case and Bezel</b>               |    |    |   |   |   |   |    |   |           |  |  |  |  |
| SS304 (Std.)                        |    |    |   |   |   |   |    |   |           |  |  |  |  |
| SS 316                              |    |    |   |   |   |   |    |   |           |  |  |  |  |
| <b>Socket</b>                       |    |    |   |   |   |   |    |   |           |  |  |  |  |
| SS 316 L (Std.)                     |    |    |   |   |   |   |    |   |           |  |  |  |  |
| <b>Movement</b>                     |    |    |   |   |   |   |    |   |           |  |  |  |  |
| SS 304 (Std.)                       |    |    |   |   |   |   |    |   |           |  |  |  |  |
| SS 316                              |    |    |   |   |   |   |    |   |           |  |  |  |  |
| <b>Pointer</b>                      |    |    |   |   |   |   |    |   |           |  |  |  |  |
| Micrometer Pointer                  |    |    |   |   |   |   |    |   |           |  |  |  |  |
| <b>Window</b>                       |    |    |   |   |   |   |    |   |           |  |  |  |  |
| Toughened Glass (Std.)              |    |    |   |   |   |   |    |   |           |  |  |  |  |
| Shatterproof Laminated Safety Glass |    |    |   |   |   |   |    |   |           |  |  |  |  |
| <b>Top Chamber</b>                  |    |    |   |   |   |   |    |   |           |  |  |  |  |
| SS 316                              |    |    |   |   |   |   |    |   |           |  |  |  |  |
| <b>Diaphragm</b>                    |    |    |   |   |   |   |    |   |           |  |  |  |  |
| SS 316 L (Std.)                     |    |    |   |   |   |   |    |   |           |  |  |  |  |
| MONEL 400                           |    |    |   |   |   |   |    |   |           |  |  |  |  |
| <b>Bottom Chamber</b>               |    |    |   |   |   |   |    |   |           |  |  |  |  |
| SS 316 (Std.)                       |    |    |   |   |   |   |    |   |           |  |  |  |  |
| MONEL 400                           |    |    |   |   |   |   |    |   |           |  |  |  |  |

| Tests and Certificates                        |
|-----------------------------------------------|
| Five point calibration certificate (Standard) |
| Material test certificates for wetted parts   |
| NACE conformity for wetted parts              |
| Helium leak test                              |
| Special requirement – please specify          |

| Optional Features                                          |
|------------------------------------------------------------|
| Over range protection of 130% of FS                        |
| Tag marking on dial                                        |
| SS tag plate                                               |
| 2" Pipe mounting bracket in SS 304                         |
| 2" Pipe mounting bracket in SS 316                         |
| Internal vacuum stop                                       |
| External zero adjustment                                   |
| Diaphragm and PTFE protection – specify type of protection |
| Special flange process connection                          |
| Special requirement – please specify                       |

Ordering example : SHE1CABAL- 0000 (Standard Code)



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